The negation map in `index_calculus.rho` (2026-10-20)

Change. `pollard_rho` used to walk on points ($A = 1$, $\approx \sqrt{\pi N/2}$ steps). It now walks on $\{X, -X\}$ by default ($A = 2$, $\approx \sqrt{\pi N/4}$), with the Wiener–Zuccherato look-ahead and a 16-point short-cycle escape by doubling, the same guards as crypto's matched reference. Look-ahead and escape additions are counted in `walk_ops`. `--rho-walk plain` (`negation=False`) is the old walk, bit-identical on 15 checked instances. Rows record `rho_walk`.

Why. A plain-walk rho is about $\sqrt{2}$ too slow as a reference and flatters IC by $\sqrt{2}$ (crypto audit X02/X03).

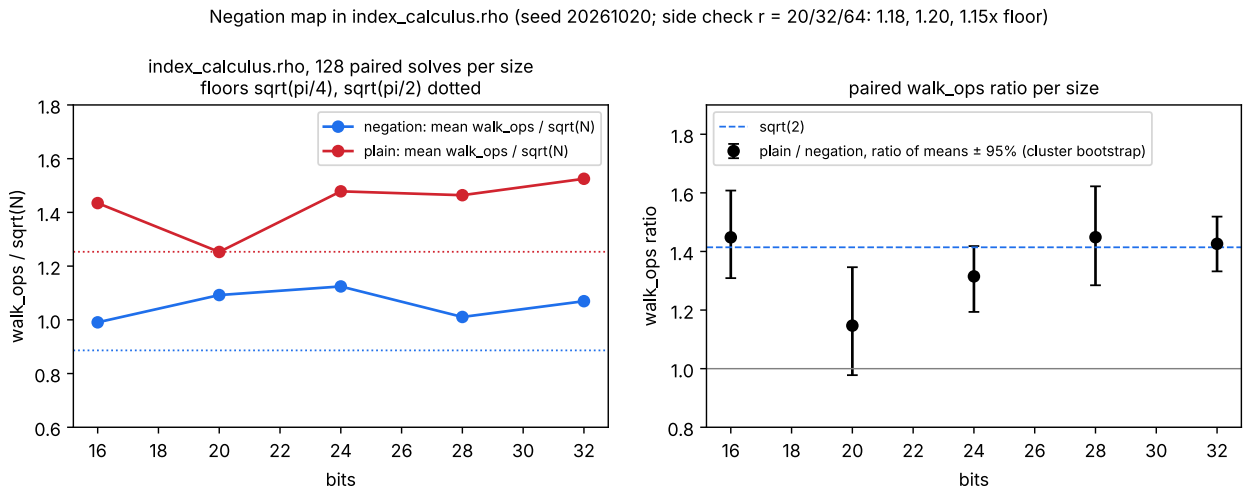


Figure 1: `measure.py`, seed 20261020: 8 prime-order curves × 16 planted logs per size, both walks on each instance with the same seed; 1,280/1,280 verified. Right: ratio of mean `walk_ops`, 95% cluster bootstrap over curves.

bits	pairs	neg / $\sqrt{N}$	/floor	plain / $\sqrt{N}$	/floor	plain/neg [95%]	wall p/n
16	128	0.991	1.118	1.435	1.145	1.448 [1.309, 1.608]	1.024
20	128	1.092	1.232	1.253	0.999	1.147 [0.978, 1.346]	0.934
24	128	1.124	1.269	1.478	1.180	1.315 [1.194, 1.418]	0.981
28	128	1.011	1.140	1.464	1.168	1.449 [1.285, 1.622]	1.101
32	128	1.069	1.207	1.525	1.217	1.426 [1.332, 1.519]	1.085
all	640	1.057	1.193	1.431	1.142	**1.353 [1.281, 1.425]**	1.081

- The pooled saving is 1.353; $\sqrt{2}$ is at the top of the interval. The negation walk is 1.19× its floor and plain is 1.14×, so about 4% of $\sqrt{2}$ is lost. The look-ahead accounts for 2.6% of additions and escapes for 0.07%.
- Table size (`jumps.py`, 384 solves each at 24–32 bits): r = 20, 32 and 64 give 1.183, 1.201 and 1.150× floor (s.e. 0.03). No clear difference; r stays 20.
- Wall time (descriptive, interpreted Python, shared VM): 1.08× pooled.
- Not changed: `harness/rho.py` and `harness/walk.py`. Companion study: aburan28/crypto `research/rho_negation_audit_20261020/` (ecbench `rho.negation` 1.371 [1.257, 1.491]; `volcano-rho` fixed the same way).